

Beliefs about Refugee Integration in Germany: A Survey Experiment on Ukrainian and Syrian Refugees

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1 Introduction

2 Literature Review and Conceptual Framework

2.1 Attitudes toward Migrants and Refugees.

2.2 Ukrainian and Syrian Refugees in Europe and Germany.

2.3 Beliefs, expectations, misperceptions, and discrimination.

2.4 Experiences, Memory, and Belief Formation.

3 Experimental Design

- i indexes respondents.
- j indexes vignette profiles.
- $g \in \{\text{Ukraine, Syria}\}$ indexes migrant groups.
- Y_{ij} denotes respondent i 's evaluation of vignette j .
- C_{ig} denotes respondent i 's contact exposure to migrant group g .
- Q_{ig} denotes respondent i 's overall contact quality with migrant group g .
- M_{ig} denotes respondent i 's media exposure regarding migrant group g .

General Notation: For inline equations, use For displayed equations, use

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or the equation environment.

3.1 Target Population and Proposed Sample

3.2 Factorial Vignette Design

The study employs a factorial vignette experiment in which respondents evaluate hypothetical migrants whose characteristics are independently varied across profiles. This approach builds on previous factorial and conjoint experiments examining how characteristics of immigrants and refugees affect evaluations by host-country populations (e.g., Hainmueller & Hopkins, 2015; Bansak et al., 2016; Czymara & Schmidt-Catran, 2016).

The vignettes describe potential migrants who currently live in Germany and differ randomly across several characteristics. The description of immigrants is based on a total of five dimensions: country of origin (Ukrainian/Syrian), reason for immigration (war

refugee/labor migrant), gender (male/female), language skills (low/good German), and education (low/high). Country and language skills describe the cultural distance. Distinguish between two levels: (1) low and (2) good German language skills. The education factor represents the human capital of the immigrants. Distinguish between a (1) low and a (2) high level of education, where: low level of education: completed secondary education but has no vocational or university degree. high level of education: has a university degree. It should also be taken into account that language skills also have an economic component. Success in the labor market depends, for example, on the ability to speak the local language. The effect size of language proficiency could exceed the effect sizes of religious affiliation and country of origin. However, it should be considered that part of the effect of language skills, in the sense of the human capital hypothesis, can also be attributed to economic considerations. (Czymara & Schmidt-Catran, 2016)

Why religious affiliation is not included in the vignette: As discussed in the literature review, previous research demonstrates that religion and perceived cultural distance can influence attitudes toward immigrants (e.g., Czymara & Schmidt-Catran, 2016). However, in this study, religious affiliation is deliberately not included as a randomized vignette attribute. The present study focuses on differences in expectations associated with migrants' country of origin, migration status, human capital, gender, and German-language proficiency. Introducing religion as an additional vignette dimension would increase the complexity of the experimental design. Moreover, explicitly providing information about religious affiliation could alter the beliefs that respondents would otherwise associate with Ukrainian and Syrian migrants based on the country of origin alone. Religion is therefore left unspecified in the vignette, allowing the country-of-origin treatment to capture respondents' overall beliefs associated with the respective national-origin group. Consequently, the estimated country-of-origin effect should be interpreted as a composite effect that may encompass perceived cultural and religious differences, rather than as an effect of nationality that is independent of religion. Respondents' own religious affiliation is nevertheless collected as a background characteristic and can be included as a control or examined in heterogeneity analyses.

Vignette wording: [Gender: Mr./Mrs.] [random letter]. came from [country of origin: Ukraine/Syria] to Germany. [He/She] came to Germany as [reason for immigration: a refugee after fleeing the war in [his/her] country / a labor migrant seeking employment opportunities in Germany]. [He/She] [skill level: has completed a university degree / has no formal vocational qualification], speaks German [language skills: well / only to a limited extent] and is currently looking for stable employment.

vignette universe: $2 \times 2 \times 2 \times 2 \times 2 = 32$ profiles

male/female $\times$ Ukraine/Syria $\times$ refugee/labor migrant $\times$ high/low skill $\times$ good/limited German = 16 profiles

Each respondent receives only a few randomly selected vignettes. The randomization allows estimation of the marginal effect of each characteristic.

1. Explained vignette attributes: *Origin_j MigrationStatus_j Gender_j Education_j GermanSkills_j*

2. Choosing particular vignette attributes: Educational attainment is included because previous vignette and conjoint evidence consistently documents more favorable evaluations of highly skilled immigrants (Hainmueller & Hiscox, 2010; Hainmueller & Hopkins, 2015; Czymara & Schmidt-Catran, 2016). Language proficiency can be connected to Czymara & Schmidt-Catran and other relevant experiments. Migration status can be justified through the refugee-versus-labor-migrant literature. Country of origin obviously comes from your substantive Ukrainian–Syrian research question.

3.3 Outcomes Measures

3.4 Respondent Questionnaire

Citing measurement sources for the respondent questionnaire: C_{ig}^{Work} workplace contact frequency C_{ig}^{Social} social contact frequency Q_{ig} contact quality

Contact quality: The notation $Q_{ig} \in \{1, \dots, 7\}$ denotes respondent i 's overall contact quality with migrant group g . Contact quality is measured on a 7-point scale, where 1 indicates very negative contact experiences and 7 indicates very positive contact experiences. The measure is adapted to distinguish between the two origin groups examined in the present study.

$$PositiveExperience_{ig} = \max(Q_{ig} - 4, 0)$$

$$NegativeExperience_{ig} = \max(4 - Q_{ig}, 0)$$

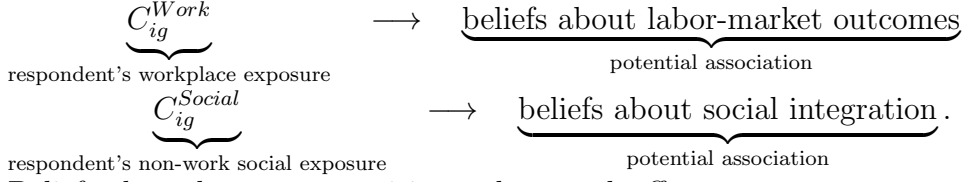
Following the intergroup-contact literature, the questionnaire distinguishes between the frequency and quality of respondents' contact with members of the two origin groups. This distinction is motivated by previous evidence showing that the quantity and valence of intergroup contact constitute conceptually distinct dimensions and may have different relationships with attitudes toward refugees (e.g., De Coninck et al., 2021). Contact is measured separately for Ukrainians and Syrians and across workplace and non-work social settings.

Media exposure: $MediaExposure_{ig}$ Media exposure is measured separately for Ukrainian and Syrian refugees. Building on work conceptualizing refugee-related media exposure as a form of indirect intergroup contact (e.g., De Coninck et al., 2021), respondents report the frequency with which they encounter relevant news and media coverage. The measure is adapted to distinguish between the two origin groups examined in the present study.

Motivated by research showing that the memorability of information and narratives may influence subsequent beliefs (Graeber et al., 2024), the questionnaire additionally elicits whether respondents can recall a particularly salient personal experience involving

members of either origin group and, if so, the valence of that experience.

Belief about migrant integration:



Beliefs about large communities and network effects:

Large community $\rightarrow$ More information/referrals $\rightarrow$ Better employment
versus

Large community $\rightarrow$ Less native interaction $\rightarrow$ Slower integration.

Construct:

$$\Delta B_i^{\text{Network}} = B_{i,S}^{\text{Network}} - B_{i,U}^{\text{Network}}$$

3.5 Randomization and Survey Procedure

4 Empirical Strategy and Identification

Identification and equations: Baseline equation:

$$Y_{ij} = \alpha + \beta_1 Ukr_j + \beta_2 Ref_j + \beta_3 Female_j + \beta_4 HighSkill_j + \beta_5 German_j + \varepsilon_{ij}.$$

Adjusted specification:

$$Y_{ij} = \alpha + \beta_1 Ukr_j + \beta_2 Ref_j + \beta_3 Female_j + \beta_4 HighSkill_j + \beta_5 German_j + \boldsymbol{\delta}' \mathbf{X}_i + \varepsilon_{ij}.$$

H4 regression with interaction term:

$$Y_{ij} = \alpha + \beta_1 Ukraine_j + \beta_2 Refugee_j + \beta_3 (Ukraine_j \times Refugee_j) + \boldsymbol{\gamma}' \mathbf{Z}_j + \varepsilon_{ij}.$$

where

$$\mathbf{Z}_j = \begin{pmatrix} Female_j \\ HighSkill_j \\ GoodGerman_j \end{pmatrix}$$

$$\mathbf{X}_i = (Age_i, Gender_i, Edu_i, Emp_i, Inc_i, Mig_i, East_i, Religion_i, Right_i)'.$$

Then,

$$Y_{ij} = \alpha + \boldsymbol{\beta}' \mathbf{Z}_j + \boldsymbol{\theta}' \mathbf{X}_i + \varepsilon_{ij},$$

where $\boldsymbol{\theta}$ is the vector of coefficients associated with the respondent controls, and $\mathbf{Z}_j$ is contains the randomized vignette characteristics and $\mathbf{X}_i$ contains respondent-level controls.

5 Limitations and Extensions

5.1 Extensions

5.2 Limitations and Identification Constraints

6 Conclusion

References

Statement of Authorship

I hereby confirm that the work presented has been performed and interpreted solely by myself except for where I explicitly identified the contrary. I assure that this work has not been presented in any other form for the fulfillment of any other degree or qualification. Ideas taken from other works in letter and in spirit are identified in every single case.

Date:

Signature: